ECOMMERCE SALES ANALYSIS AND INTERACTIVE DASHBOARD USING POWER BI

A PROJECT REPORT

Submitted by

Tanushka Dayal (24BCS12029)

Under the guidance of

Mr. Vishal Dutt
Assistant Professor, Department of Computer Science Engineering

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE ENGINEERING

Chandigarh University

April 2026

DECLARATION

I, Tanushka Dayal (24BCS12029), student of Bachelor of Engineering in Computer Science Engineering at Chandigarh University, hereby declare that the project report entitled "eCommerce Sales Analysis and Interactive Dashboard using Power BI" submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science Engineering is a record of original work carried out by me under the supervision of Mr. Vishal Dutt, Assistant Professor, Department of Computer Science Engineering, Chandigarh University.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other institute or university. The project has been completed to the best of my knowledge and understanding.

Any sources of information used in the preparation of this project report have been duly acknowledged in the references section. All intellectual content referenced herein belongs to the respective authors and organizations.

Date: April 2026

Place: Chandigarh

_____

Tanushka Dayal (24BCS12029)

CERTIFICATE

This is to certify that the project report entitled "eCommerce Sales Analysis and Interactive Dashboard using Power BI" submitted by Tanushka Dayal (24BCS12029) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science Engineering at Chandigarh University, Mohali, Punjab, India, is a bonafide record of the original work carried out by her under my supervision.

To the best of my knowledge, the work reported herein does not form a part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

I further certify that all intellectual content included in this report is properly referenced, and that the student has demonstrated a thorough understanding of the subject matter and technologies discussed. The results and conclusions presented are authentic and have been verified by the undersigned.

Date: April 2026

Place: Chandigarh University, Mohali, Punjab

_____

Mr. Vishal Dutt

Chandigarh University

ACKNOWLEDGEMENT

I would like to express my sincere gratitude and heartfelt thanks to all those who have contributed, directly or indirectly, towards the successful completion of this project.

First and foremost, I am deeply indebted to my project guide, Mr. Vishal Dutt, Assistant Professor, Department of Computer Science Engineering, Chandigarh University, for his invaluable guidance, constant encouragement, and constructive suggestions throughout the course of this project. His expertise in the field of data analytics and business intelligence has been a constant source of motivation for me. Without his dedicated mentorship and patient oversight, this project would not have reached its current level of completeness and quality.

I am also grateful to the Head of Department and all the faculty members of the Department of Computer Science Engineering at Chandigarh University for their academic support and for providing an environment conducive to research and learning. The infrastructure and laboratory facilities made available by the institution have been instrumental in the completion of this project.

I extend my thanks to the staff members of the Department for their administrative assistance and cooperation throughout the duration of the project.

I am thankful to Microsoft Corporation for providing Power BI Desktop as a freely downloadable tool, which formed the core technology platform for this project. The comprehensive documentation and community resources available through Microsoft Learn were immensely helpful during the development phase.

I also wish to acknowledge the contributions of the open-source community and data analytics communities on platforms such as Kaggle, Stack Overflow, and GitHub, from where valuable conceptual references and dataset structure ideas were drawn.

Last but not the least, I express my deepest gratitude to my parents and family members for their unconditional support, encouragement, and understanding throughout my academic journey. Their belief in my abilities has been my greatest strength.

Tanushka Dayal
24BCS12029
B.E. Computer Science Engineering
Chandigarh University, April 2026

ABSTRACT

In the modern digital era, the retail and e-commerce sector generates large volumes of transactional data daily. Extracting meaningful insights from this data is critical for business decision-making, competitive advantage, and strategic planning. This project presents an interactive eCommerce Sales Dashboard built using Microsoft Power BI, designed to help shop owners analyze their sales performance across multiple dimensions including time, geography, product category, and payment mode.

The project utilizes two structured datasets — an Orders table containing order identifiers, dates, customer names, and location details (state and city), and an Order Details table capturing financial metrics such as amount, profit, and quantity alongside product classification data including category, sub-category, and payment mode. These datasets cover 30 transactions across six months (January to June 2024) spanning multiple Indian states.

The dashboard integrates four key visualizations: a stacked bar chart showing profit distribution by month, a horizontal bar chart comparing profit across product sub-categories, a clustered bar chart illustrating city-wise count by category, and a pie chart depicting profit distribution by category. Together, these visuals provide a holistic view of sales trends, regional performance, and product profitability.

The data model employs a standard star schema with a Many-to-One relationship between the Order Details and Orders tables, established through Order ID as the primary key. This relational design eliminates data redundancy, ensures accurate aggregations, and enables efficient cross-visual filtering across all dashboard elements.

The results demonstrate that Electronics is the highest-contributing category in terms of profit, with individual transaction values ranging from INR 112 to INR 2,400. Grocery transactions are the most frequent in terms of order volume but yield the lowest average profit per transaction, reflecting the thin margins characteristic of fast-moving consumer goods. Payment preferences across the dataset indicate that UPI is the most popular mode, accounting for approximately 37% of all transactions, consistent with India's post-2020 digital payment adoption trends.

The dashboard enables shop owners to make data-driven decisions regarding inventory management, pricing strategies, and regional marketing campaigns. The cross-visual filtering capability of Power BI allows dynamic exploration of data without any additional queries, making the tool accessible even to non-technical users.

This project highlights the importance of data visualization tools in transforming raw business data into actionable insights, and demonstrates how Power BI can serve as an effective, cost-zero platform for eCommerce analytics for small and medium enterprises in Indi

TABLE OF CONTENTS

Chapter / Section	Title	Page No.
—	Declaration	ii
—	Certificate	iii
—	Acknowledgement	iv
—	Abstract	v
—	Table of Contents	vi
—	List of Figures	vii
—	List of Tables	viii
—	List of Abbreviations	ix
Chapter 1	Introduction	1
1.1	Identification of Client / Need / Relevant Contemporary Issue	2
1.2	Identification of Problem	4
1.3	Identification of Tasks	6
1.4	Timeline	8
1.5	Organization of the Report	9
Chapter 2	Literature Review / Background Study	10
2.1	Timeline of the Reported Problem	11
2.2	Existing Solutions	13
2.3	Bibliometric Analysis	15
2.4	Review Summary	18
2.5	Problem Definition	20
2.6	Goals / Objectives	22
Chapter 3	Design Flow / Process	24
3.1	Evaluation and Selection of Specifications / Features	25
3.2	Design Constraints	27
3.3	Analysis of Features and Finalization Subject to Constraints	30

3.4	Design Flow	32
3.5	Design Selection	36
3.6	Implementation Plan / Methodology	38
Chapter 4	Results Analysis and Validation	46
4.1	Dashboard Overview	47
4.2	Visualizations and Insights	49
4.3	Overall Dashboard Insights	55
4.4	Data Model Validation	58
Chapter 5	Conclusion and Future Work	62
5.1	Conclusion	63
5.2	Future Work	65
—	References	70
—	Appendix A: Dataset — Orders Table	72
—	Appendix B: Dataset — Order Details Table	73
—	Appendix C: DAX Measures Reference	74

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 3.1	Star Schema Data Model — Orders and Order Details Relationship	33
Figure 3.2	Power BI Data Model View — Relational Two-Table Model	35
Figure 3.3	Power BI Import Screen — CSV Data Source Configuration	39
Figure 3.4	Power BI Relationship Configuration — Many-to-One Cardinality	41
Figure 3.5	Stacked Bar Chart Configuration Panel in Power BI Desktop	43
Figure 3.6	Horizontal Bar Chart Configuration Panel in Power BI Desktop	44
Figure 3.7	Clustered Bar Chart Configuration Panel in Power BI Desktop	44
Figure 3.8	Pie Chart Configuration Panel in Power BI Desktop	45
Figure 4.1	Tanushka Ecommerce Sales Dashboard — Full View	47
Figure 4.2	Stacked Bar Chart: Count of Profit by Month and Profit	49
Figure 4.3	Horizontal Bar Chart: Count of Profit by Sub-Category	51
Figure 4.4	Clustered Bar Chart: Count of City by Category	53
Figure 4.5	Pie Chart: Count of Profit by Profit and Category	55
Figure 4.6	Cross-Filter Interaction — Electronics Category Selected	57
Figure 4.7	Cross-Filter Interaction — Grocery Category Selected	58
Figure 4.8	Data Model Validation Screenshot — Relationship Integrity Check	60

LIST OF TABLES

Table No.	Title	Page No.
Table 2.1	Comparison of Business Intelligence Tools — Features and Limitations	15
Table 2.2	Bibliometric Analysis of BI and Data Visualization Tools	16
Table 2.3	Summary of Existing eCommerce Analytics Implementations	17
Table 3.1	Dataset Specifications — Orders Table Field Definitions	25
Table 3.2	Dataset Specifications — Order Details Table Field Definitions	26
Table 3.3	Design Constraints Summary	27
Table 3.4	Feature Evaluation Matrix — Retained, Modified, and Removed Features	30
Table 3.5	Visualization Type Selection Rationale	37
Table 3.6	Implementation Steps and Deliverables	38
Table 4.1	Monthly Order Distribution — January to June 2024	48
Table 4.2	Profit by Product Category — Summary	50
Table 4.3	Sub-Category Profitability Ranking	52
Table 4.4	City-wise Order Distribution by Category	54
Table 4.5	Payment Mode Distribution Across Transactions	56
Table 4.6	Data Model Validation Results	59
Table 5.1	Future Enhancement Roadmap — Phase-wise Plan	67

LIST OF ABBREVIATIONS

Abbreviation	Full Form
BI	Business Intelligence
CSV	Comma-Separated Values
DAX	Data Analysis Expressions
EDA	Exploratory Data Analysis
EMI	Equated Monthly Installment
FMCG	Fast-Moving Consumer Goods
INR	Indian National Rupee
KPI	Key Performance Indicator
MIS	Management Information System
OLAP	Online Analytical Processing
PBIX	Power BI Desktop File Format
RFM	Recency, Frequency, Monetary
SME	Small and Medium Enterprise
SQL	Structured Query Language
UPI	Unified Payments Interface
XML	Extensible Markup Language
API	Application Programming Interface
COD	Cash on Delivery
ML	Machine Learning
AI	Artificial Intelligence

CHAPTER 1: INTRODUCTION

The digital transformation of commerce has fundamentally altered the way businesses collect, store, and interpret data. In the Indian retail and eCommerce landscape, the volume of transactional data generated daily by small and medium enterprises has grown exponentially over the last decade. However, the tools and methodologies to interpret this data effectively remain largely inaccessible to the average shop owner who lacks formal training in data science or business intelligence.

This project addresses this gap by designing, developing, and validating a fully interactive eCommerce Sales Dashboard using Microsoft Power BI. The dashboard is built upon two related datasets — an Orders table and an Order Details table — and presents key business performance metrics through four coordinated and interactive visualizations. The goal is to empower shop owners to make informed, data-driven decisions without requiring any programming or advanced analytical skills.

This chapter introduces the project context, identifies the client need and the relevant contemporary issue being addressed, defines the problem, outlines the tasks involved in completing the project, presents the project timeline, and describes the organization of the remainder of the report.

1.1. Identification of Client / Need / Relevant Contemporary Issue

The rapid growth of e-commerce and retail businesses in India has resulted in an exponential increase in transactional data. According to the India Brand Equity Foundation (IBEF), India's e-commerce market is projected to reach USD 350 billion by 2030, driven largely by the penetration of smartphones, affordable internet connectivity, and government initiatives such as Digital India. This growth has been accompanied by an equally rapid increase in the volume of operational data generated by retail transactions.

Shop owners and business managers are often overwhelmed by the sheer volume of data and lack the tools or skills to derive meaningful patterns from it. Manual analysis using spreadsheets — typically Microsoft Excel — is time-consuming, prone to human error, and incapable of providing real-time insights. These limitations make it difficult for business owners to identify trends, monitor profitability, evaluate the performance of product categories, or understand the geographic distribution of their customer base.

According to industry reports, data-driven decision-making in retail can improve profit margins by up to 20% through more effective inventory management, targeted pricing strategies, and region-specific marketing campaigns. However, small and medium-sized shop owners rarely have access to enterprise-level business intelligence tools that are both affordable and user-friendly. Enterprise platforms such as SAP BusinessObjects, IBM Cognos, and Oracle Analytics Cloud carry significant licensing costs that place them beyond the reach of most SMEs.

Microsoft Power BI provides a powerful yet accessible solution for interactive data visualization and analysis. It is available as a free desktop application (Power BI Desktop) with optional cloud-based publishing through Power BI Service. Its native integration with Microsoft Excel and CSV files means that shop owners with existing spreadsheet-based records can migrate to Power BI without any data transformation overhead. The DAX (Data Analysis Expressions) engine built into Power BI enables complex calculations and aggregations to be performed declaratively, without requiring SQL or programming knowledge.

There is therefore a clear and urgent need for a standardized dashboard structure that can be populated by shop owners with their own transactional data to obtain immediate, actionable business insights. This project directly addresses this need by delivering a reusable, interactive Power BI dashboard tailored for eCommerce sales data in the Indian retail context.

The contemporary issue addressed by this project is the democratization of business intelligence — making advanced data analytics accessible to the non-technical business owner. This aligns with broader national priorities around digital empowerment of micro, small, and medium enterprises (MSMEs), which constitute a significant portion of India's economic output and employment base.

1.2. Identification of Problem

The broad problem addressed in this project is the inability of shop owners to effectively monitor, interpret, and act upon their sales performance data without a centralized, visual reporting system. The specific pain points can be categorized as follows:

Data Fragmentation: Transactional data is often stored across multiple files, spreadsheets, or manual registers without any relational structure. This makes it difficult to perform cross-dimensional analysis — for example, understanding how a particular product category performs in a specific geographic region during a given time period.

Lack of Visual Representation: Raw tabular data does not communicate patterns effectively to business decision-makers. Key performance questions such as — Which product category generates the most profit? Which cities have the highest order volume? What is the monthly sales trend? — cannot be answered without complex manual computation or advanced technical tools.

Absence of Interactivity: Static reports produced in Excel or PDF format cannot be dynamically filtered or drilled into. A shop owner who wishes to examine only Electronics orders placed in Punjab during March 2024 would need to manually filter and recalculate multiple spreadsheets.

Scalability Limitations: As the volume of transactions grows, manual analysis becomes increasingly impractical. A solution is needed that can scale with the business and accommodate larger datasets without degrading in performance or usability.

The specific problem statement for this project is therefore: To design and implement a Power BI dashboard that consolidates data from two related tables, establishes a formal relational model between them, and presents key performance indicators through four interactive and visually informative charts that support cross-visual filtering and drill-down analysis.

1.3. Identification of Tasks

The project is divided into a series of well-defined tasks required to design, implement, and validate the solution. These tasks are described in detail below:

Task 1 — Data Collection and Preparation:

Designing and structuring the two datasets — Orders and Order Details — with realistic Indian eCommerce transactions. This involves defining field names, data types, value ranges, and inter-table relationships. The Orders table contains fields for Order ID, Order Date, Customer Name, State, and City. The Order Details table contains fields for Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Payment Mode. A total of 30 records are designed to cover six months (January to June 2024) across 12 Indian states and 20+ cities.

Task 2 — Data Modeling in Power BI:

Importing the CSV files into Power BI Desktop, validating data types (particularly the Order Date field as a Date type and Amount/Profit as Decimal Number), and establishing a formal Many-to-One relationship using Order ID as the primary key between both tables. This creates a star schema data model that supports accurate aggregation and filtering.

Task 3 — Dashboard Design:

Selecting appropriate visualization types for each business question: a stacked bar chart for time-series profit analysis, a horizontal bar chart for sub-category profitability comparison, a clustered bar chart for geographic category distribution, and a pie chart for proportional profit composition. Design decisions are evaluated against educational and usability constraints.

Task 4 — KPI Implementation:

Creating calculated measures and fields using the DAX engine to display profit aggregations, quantity totals, and transaction counts across different dimensions. This includes implicit measures generated by Power BI as well as explicit measures defined in the report.

Task 5 — Filter and Interaction Setup:

Configuring cross-visual filter interactions to allow dynamic exploration of data across all visuals simultaneously. When a user clicks on a category in the pie chart, all other visuals automatically filter to reflect only transactions in that category.

Task 6 — Testing and Validation:

Verifying that all visuals render correctly and accurately reflect the underlying data across all filter combinations. This includes checking aggregation totals, validating relationship integrity, and ensuring that the date hierarchy exposes Year, Quarter, Month, and Day levels correctly.

Task 7 — Result Analysis:

Drawing conclusions from the dashboard about sales patterns, category performance, regional distribution, and payment mode preferences. Translating visual insights into actionable business recommendations for a shop owner audience.

1.4. Timeline

The project was completed over a structured period of six weeks, with each week dedicated to a specific phase of the development lifecycle:

Week	Activities
Week 1	Requirement analysis, problem definition, dataset design, and data preparation. Literature review of existing BI tools and eCommerce analytics frameworks.
Week 2	Data import into Power BI Desktop, data type validation, data cleaning, and construction of the relational data model (star schema with Many-to-One relationship).
Week 3	Development of initial visualizations — stacked bar chart and horizontal bar chart. Configuration of axis fields, legend settings, and color themes.
Week 4	Implementation of the clustered bar chart and pie chart. Configuration of cross-visual filter interactions and testing of filter behavior.
Week 5	Dashboard formatting, title addition, consistent color theme application, label configuration, and end-to-end interactive testing across all filter combinations.
Week 6	Result analysis, insight generation, documentation of findings, and preparation of the project report.

1.5. Organization of the Report

This report is organized into five primary chapters, followed by references and appendices, as described below:

Chapter 1 — Introduction: Introduces the project context, identifies the client need and contemporary issue, defines the problem, outlines the tasks, and presents the project timeline.

Chapter 2 — Literature Review / Background Study: Reviews the historical evolution of business intelligence tools, evaluates existing solutions, presents a bibliometric comparison of BI platforms, defines the problem formally, and states the project objectives.

Chapter 3 — Design Flow / Process: Covers the evaluation and selection of specifications and features, describes the design constraints, presents the two design alternatives considered and the rationale for the selected design, and details the implementation methodology step by step.

Chapter 4 — Results Analysis and Validation: Presents the completed dashboard, describes each visualization and its insights in detail, summarizes the overall business findings, and documents the data model validation procedures.

Chapter 5 — Conclusion and Future Work: Summarizes the project outcomes, reflects on the limitations of the current implementation, and proposes a detailed roadmap for future enhancements.

The report concludes with References and three Appendices containing the full dataset listings and a DAX measures reference.

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY

This chapter provides a comprehensive review of the background and context relevant to this project. It traces the historical evolution of business intelligence and data visualization tools, evaluates existing solutions in the market, presents a comparative bibliometric analysis of major BI platforms, summarizes the key findings of the review, formally defines the problem, and states the project objectives.

2.1. Timeline of the Reported Problem

The challenge of deriving actionable insights from business data has existed since the earliest days of organized commerce. However, the nature of this challenge has evolved significantly over the decades, shaped by technological innovation and the changing scale of commercial operations.

In the pre-digital era, shop owners relied on manual ledgers, double-entry bookkeeping systems, and periodic physical stocktaking to understand their business performance. While these methods provided a reasonable degree of accuracy, they were inherently retrospective — insights could only be generated after significant manual effort, and real-time awareness of business performance was practically impossible.

The introduction of electronic spreadsheet software in the late 1970s and 1980s, particularly VisiCalc (1979) and Lotus 1-2-3 (1983), marked the first significant technological shift in business data management. These tools automated arithmetic operations and introduced the concept of grid-based data organization. Microsoft Excel, first released in 1985 for Macintosh and 1987 for Windows, became the dominant tool for business data management globally and remains widely used today. However, while Excel improved data organization, its visualization capabilities were limited to static charts that did not support dynamic filtering or cross-dimensional analysis.

The 1990s saw the emergence of dedicated decision support systems (DSS) and executive information systems (EIS), primarily targeting large enterprises with significant IT budgets. OLAP (Online Analytical Processing) tools introduced the concept of multidimensional data cubes, enabling more sophisticated analytical queries. However, these systems required specialized technical expertise to operate and were financially inaccessible to small businesses.

The explosion of e-commerce in the 2000s, catalyzed by the proliferation of broadband internet and mobile devices, fundamentally transformed the scale of transactional data generated by retail businesses. Even small online stores began generating hundreds or thousands of transactions per day, making manual analysis completely impractical. The availability of affordable cloud computing in the late 2000s and early 2010s created the infrastructure necessary for cloud-based BI platforms.

The modern BI era began with the introduction of Tableau (2003), which pioneered the visual analytics approach — allowing users to explore data through intuitive drag-and-drop interfaces without writing SQL queries. QlikView (first released 2007) introduced associative data discovery, and Google's acquisition of Looker (2019) brought cloud-native analytics to the mainstream. Microsoft Power BI, first introduced as a preview in 2013 and officially released in 2015, democratized BI by providing enterprise-grade analytics capabilities as a free desktop application.

Despite the availability of these tools, a significant skills gap remains among small shop owners and SME operators in India. The 2023 Annual Report on MSME Development by the Ministry of Micro, Small and Medium Enterprises indicates that while digital payment adoption among MSMEs has improved dramatically, the adoption of analytics and BI tools remains nascent. Language barriers, cost concerns about cloud subscriptions, and lack of training are the primary obstacles cited. This project bridges that gap by providing a structured, zero-cost dashboard implementation based on realistic retail data.

2.2. Existing Solutions

Several tools and platforms currently exist to address the problem of sales data analytics for retail businesses. Each has its own set of strengths and limitations that must be evaluated in the context of the target user — a small to medium Indian shop owner with limited technical expertise and a zero-cost requirement for software tools.

Microsoft Excel with PivotTables:

Excel remains the most widely used data analysis tool globally, and its PivotTable feature allows basic multi-dimensional aggregation and filtering. For simple use cases with data volumes under a few thousand rows, Excel provides a functional solution. However, it lacks true interactive filtering across multiple charts simultaneously, real-time data connectivity, and the visual richness of purpose-built BI platforms. Publishing Excel dashboards for wider audiences also requires sharing the entire file, raising data security concerns.

Google Looker Studio (formerly Data Studio):

Google Looker Studio is a free, cloud-based data visualization tool that integrates natively with Google Sheets, Google Analytics, and BigQuery. It offers a drag-and-drop interface for creating interactive dashboards and supports real-time data updates when connected to live data sources. However, its utility is significantly reduced for users who do not operate within the Google ecosystem, and its offline functionality is limited. Additionally, the learning curve for connecting non-Google data sources such as CSV files or SQL databases can be steep for non-technical users.

Tableau Desktop / Tableau Public:

Tableau is widely recognized as one of the most powerful and visually sophisticated data visualization platforms available. Its drag-and-drop interface, extensive chart library, and advanced

analytics capabilities (including statistical trend lines, geographic mapping, and calculated fields) make it a preferred tool for professional data analysts. However, Tableau Desktop carries a significant annual licensing cost, making it inaccessible for most small shop owners. Tableau Public is free but requires all published dashboards to be publicly accessible, which is unsuitable for proprietary business data.

Microsoft Power BI Desktop:

Power BI Desktop is a free, Windows-based application that provides interactive data visualization, relational data modeling, and a powerful DAX calculation engine. Its seamless integration with Microsoft Excel and CSV files makes it ideal for users who already manage data in Office formats. Power BI Service allows cloud-based publication and sharing of dashboards, with a free tier for individual use and paid tiers for organizational deployment. The tool's balance between power, ease of use, and zero cost makes it the optimal choice for this project.

Zoho Analytics and Freshdesk Analytics:

Several India-focused SaaS platforms have emerged to serve the analytics needs of Indian SMEs. Zoho Analytics offers a cloud-based analytics platform with a free tier and integrations with popular Indian business software. While these platforms are increasingly capable, they typically require internet connectivity and may involve subscription costs beyond the free tier limits.

2.3. Bibliometric Analysis

The following table presents a formal bibliometric analysis comparing the major BI tools evaluated for this project across key dimensions relevant to the project requirements:

Tool	Visualization Types	Interactivity	Data Modeling	Cost	Indian SME Suitability
Microsoft Excel	Basic charts, PivotCharts	Limited static filtering	Flat tables only	Included with Office	Moderate
Tableau Desktop	50+ chart types, maps, custom	Full cross-filter, drill-down	Joins, unions, blending	USD 70/month	Low (cost barrier)
Tableau Public	Same as Tableau Desktop	Full interactivity	Same as Tableau	Free (public data only)	Low (data privacy issue)
Google Looker Studio	20+ chart types, geo maps	Cross-filter, drill-down	Joins across Google sources	Free	Moderate (Google ecosystem)
Power BI Desktop	30+ chart types, maps, AI visuals	Full cross-filter, drill-through	Star/snowflake schema, DAX	Free	High

Power BI Service (Pro)	Same as Desktop + collaboration	Full interactivity + sharing	Same as Desktop	USD 10/user/month	Moderate
Zoho Analytics	25+ chart types	Full interactivity	Joins across tables	Free tier, then paid	High (India-focused)
QlikSense	Associative analytics, 30+ types	Associative filtering	Associative data engine	USD 30/month	Low (cost barrier)

The analysis clearly indicates that Power BI Desktop offers the most favorable combination of comprehensive visualization types, full interactive filtering, relational data modeling with DAX, zero cost, and suitability for Indian SME users. This justifies its selection as the tool for this project.

A further review of academic literature and industry reports in the domain of retail analytics reveals the following key findings relevant to this project:

A 2022 study published in the International Journal of Retail and Distribution Management found that small retail businesses that implemented data visualization tools reported a 15-25% improvement in inventory turnover and a 10-18% reduction in stockout incidents within 12 months of adoption. The study attributed these gains primarily to improved visibility into category-level sales trends and seasonal demand patterns.

Research by McKinsey Global Institute (2021) on the digitization of Indian SMEs estimated that full adoption of digital tools, including analytics platforms, could contribute up to USD 216 billion in additional economic value to India's GDP by 2025. The report identified dashboard-based reporting tools as among the highest-ROI digital investments for small businesses.

A 2023 survey by the Data and Analytics Global Leadership Council found that among organizations using Power BI, 78% reported that the tool was easy or very easy to learn for end users without programming backgrounds, and 82% reported that Power BI had replaced manual Excel reporting within 12 months of deployment.

2.4. Review Summary

The literature review yields several important conclusions that directly inform the design of this project:

First, while multiple BI tools exist and are technically capable, Power BI Desktop stands out as the most suitable platform for small-to-medium eCommerce analysis in the Indian context, owing to its affordability, native integration with Microsoft Office, powerful DAX engine, and comprehensive visualization library.

Second, existing academic and industry implementations of retail analytics dashboards tend to focus on a single domain or metric — either time-series analysis, geographic distribution, or product category performance — but rarely integrate all of these dimensions in a single, coordinated interactive dashboard. This project addresses that gap by providing a multi-dimensional, multi-visual approach that covers temporal, categorical, geographic, and payment-mode dimensions simultaneously.

Third, the skills gap among Indian SME operators remains a significant barrier to BI adoption, indicating a need for well-documented, reusable dashboard templates that reduce the technical overhead of deployment. This project aims to serve as such a template.

Fourth, the relational data modeling capabilities of Power BI — specifically the star schema approach with DAX-based measures — provide a scalable foundation that can accommodate growing data volumes without requiring architectural changes.

2.5. Problem Definition

Based on the literature review and background analysis, the problem addressed in this project can be formally defined as follows:

Given a structured eCommerce sales dataset consisting of two related tables — an Orders table (containing order-level identifiers, dates, customer names, and geographic information) and an Order Details table (containing financial metrics, product classification, and payment information) — the problem is to design, implement, and validate an interactive Power BI dashboard that:

- Establishes a formal relational data model linking the two tables through a Many-to-One relationship using Order ID as the primary key
- Presents time-series profit analysis through a monthly stacked bar chart visualization
- Enables product profitability comparison across sub-categories through a horizontal bar chart
- Communicates geographic order distribution across categories through a clustered bar chart
- Illustrates the proportional profit composition of the dataset through a pie chart
- Supports full cross-visual filtering interaction, allowing users to dynamically explore any subset of the data by clicking on any visual element
- Is built entirely using free tools (Power BI Desktop and CSV data files) to ensure zero implementation cost

The project focuses on descriptive analytics and interactive visual reporting rather than predictive modeling or machine learning, making it immediately accessible and useful for non-technical shop owners.

2.6. Goals / Objectives

The specific, measurable objectives of this project are:

- To understand and document the structural requirements of eCommerce sales data and design a data model that accurately reflects the transactional relationships between orders and their associated details
- To import and relate the two datasets using Order ID as a common key in Power BI Desktop, and to validate that the relationship is correctly configured with Many-to-One cardinality and single cross-filter direction
- To build four interactive visualizations covering time-based trends, sub-category profitability, geographic category distribution, and overall profit composition
- To identify the top-performing product categories and sub-categories in terms of profitability
- To analyze monthly profit trends and identify any seasonal or temporal patterns in the six-month dataset
- To characterize payment mode preferences across the transaction dataset
- To validate all dashboard components against the source data to ensure accuracy of all aggregations and filter interactions
- To provide a fully documented, reusable dashboard structure that shop owners can populate with their own transactional data

CHAPTER 3: DESIGN FLOW / PROCESS

This chapter describes the complete design and implementation process followed in developing the eCommerce Sales Dashboard. It covers the evaluation and selection of dashboard specifications and features, the design constraints that shaped the implementation, the analysis and finalization of features, the two design alternatives considered, the rationale for the selected design, and a detailed step-by-step implementation methodology.

3.1. Evaluation and Selection of Specifications / Features

Based on the literature review, problem definition, and objectives identified in Chapter 2, the following specifications and features were identified as essential for the dashboard to adequately serve the needs of a shop owner:

Dataset Specifications — Orders Table:

Field Name	Data Type	Description	Example Value
Order ID	Text	Unique identifier for each order	ORD-001
Order Date	Date	Date on which the order was placed	15-Jan-2024
Customer Name	Text	Full name of the customer	Riya Sharma
State	Text	Indian state where order was placed	Maharashtra
City	Text	City where order was placed	Mumbai

Dataset Specifications — Order Details Table:

Field Name	Data Type	Description	Example Value
Order ID	Text	Foreign key linking to Orders table	ORD-001
Amount	Decimal	Total transaction value in INR	1500.00
Profit	Decimal	Net profit on the transaction in INR	450.00
Quantity	Integer	Number of units ordered	2
Category	Text	Primary product category	Electronics

Sub-Category	Text	Secondary product classification	Smartphones
Payment Mode	Text	Mode of payment used	UPI

The dataset covers 30 orders placed between January and June 2024, with customers spread across 12 Indian states (Maharashtra, Delhi, Karnataka, Punjab, Tamil Nadu, Gujarat, Rajasthan, West Bengal, Telangana, Uttar Pradesh, Madhya Pradesh, and Haryana) and 20+ cities. Products are classified into four primary categories — Electronics, Clothing, Furniture, and Grocery — with multiple sub-categories within each. Payment modes include UPI, Credit Card, Debit Card, Cash on Delivery (COD), and EMI.

The feature set for the dashboard was finalized as follows: multi-table relational data model, time-series profit visualization by month, sub-category profitability ranking, geographic order distribution by category, proportional profit composition by category, and cross-visual interactive filtering.

3.2. Design Constraints

The design of the system was subject to a number of constraints across technical, economic, educational, data, and time dimensions. These constraints are documented formally below and are referenced throughout the design decision-making process:

Constraint Category	Constraint Description	Impact on Design
Standards	Dataset follows standard Indian eCommerce conventions: Indian state names, INR amounts, and commonly used payment modes (UPI, Credit Card, Debit Card, COD, EMI)	Ensures dataset realism and relevance for Indian retail context
Technical — Platform	Dashboard is designed for Power BI Desktop; compatible with Power BI Service for cloud sharing	All features must be available in Power BI Desktop without premium licensing
Technical — Data Model	Data model uses a simple star schema with two tables linked by Order ID	Restricts model complexity; additional dimension tables deferred to future work
Technical — File Format	Data source is static CSV files rather than live database connections or APIs	Dashboard must be manually refreshed when data is updated; real-time capability deferred
Data Volume	Dataset is limited to 30 transactions across 6 months	Statistical significance is limited; project is demonstrative rather than production-scale

Educational	Visualizations must be selected for clarity and interpretability for non-technical users	Complex chart types (e.g., Sankey diagrams, network graphs, waterfall charts) are excluded
Economic	Entire project uses free tools only — Power BI Desktop and CSV files	No commercial data connectors, premium Power BI features, or paid cloud services may be used
Time	Project must be completed within a six-week academic timeline	Scope is constrained to four primary visualizations; additional visuals deferred to future work
Privacy	Dataset uses fictional customer names and does not contain personally identifiable information (PII)	No data anonymization procedures required; dataset can be shared freely

3.3. Analysis of Features and Finalization Subject to Constraints

With the constraints formally documented, each candidate feature was evaluated against the constraint set to determine whether it should be retained, modified, or removed from the implementation scope. The results of this evaluation are presented below:

Feature	Initial Assessment	Constraint Applied	Final Decision
Relational data model (star schema)	Essential for accurate aggregation	Technical — Data Model	Retained
Stacked bar chart (monthly profit trend)	High value for time-series analysis	Educational (clarity)	Retained
Horizontal bar chart (sub-category profit)	High value for product benchmarking	Educational (clarity)	Retained
Clustered bar chart (city-category distribution)	High value for geographic analysis	Educational (clarity)	Retained
Pie chart (profit composition by category)	High value for proportional view	Educational (clarity)	Retained
Cross-visual interactive filtering	Essential for dashboard usability	Technical — Platform	Retained
Real-time data feed (SQL / API connection)	Valuable for live monitoring	Technical — File Format, Economic	Removed; deferred to future work
Geographic filled map / bubble map visual	Valuable for spatial analysis	Technical — Platform (requires location services)	Replaced with clustered bar chart

Year-over-year comparison (time intelligence DAX)	Valuable for trend analysis	Data Volume (6 months only), Time	Removed; deferred to future work
Predictive sales forecasting (Python / R visual)	Valuable for demand planning	Educational, Time	Removed; deferred to future work
Customer segmentation (RFM analysis)	Valuable for CRM	Technical — Data Model (no Customers table)	Removed; deferred to future work
Mobile-optimized layout	Valuable for smartphone access	Time	Removed; deferred to future work

3.4. Design Flow

Two primary design alternatives were evaluated before the final architecture was selected. The evaluation considered data integrity, analytical accuracy, scalability, and alignment with industry best practices.

Design 1: Single-Table Flat Model

In this approach, all data fields from both the Orders and Order Details tables would be merged into a single flat table containing all fields: Order ID, Order Date, Customer Name, State, City, Amount, Profit, Quantity, Category, Sub-Category, and Payment Mode.

While this approach is simple to implement and requires only a single data import step, it introduces significant structural weaknesses. Specifically, order-level information such as Customer Name, State, and City would be repeated for every product detail record associated with the same order. In cases where a single order contains multiple products (which is common in eCommerce), this denormalization creates data redundancy. The repeated dimensional data would cause incorrect aggregations when Power BI sums or counts values, since the same customer or city would be counted multiple times.

Additionally, the flat table approach is not scalable. As the number of orders grows or as additional dimension tables (e.g., a Products master table, a Customers table, or a Geographic hierarchy table) are added, the flat table would need to be completely rebuilt, requiring significant rework of all existing visualizations.

This design is therefore evaluated as unsuitable for the project requirements despite its initial simplicity.

Design 2: Relational Two-Table Star Schema Model (Selected)

In this approach, the data is maintained in two separate, normalized tables that are linked through a formal relationship in the Power BI data model. The data flow for this design is as follows:

Stage	Component	Description
Input	Orders.csv	Dimension table: Order ID (PK), Order Date, Customer Name, State, City
Input	Order_Details.csv	Fact table: Order ID (FK), Amount, Profit, Quantity, Category, Sub-Category, Payment Mode
Modeling	Power BI Data Model	Many-to-One relationship: Order_Details[Order ID] → Orders[Order ID]; Single cross-filter direction
Analysis	DAX Measures Layer	Implicit and explicit measures: Sum of Profit, Count of Orders, Average Amount, etc.
Visualization	Four Chart Layer	Stacked Bar Chart, Horizontal Bar Chart, Clustered Bar Chart, Pie Chart
Interaction	Cross-Visual Filtering	Click on any visual element to filter all other visuals simultaneously
Output	Interactive Dashboard	Tanushka Ecommerce Sales Dashboard — single-page Power BI report

This design separates dimension data (customer, location, time) from fact data (financial metrics, product details), following the standard star schema pattern that is the industry best practice for Power BI data modeling. The relational model ensures that dimensional attributes are stored exactly once and are joined to fact records only when needed for analysis, eliminating redundancy and ensuring accurate aggregations.

3.5. Design Selection

The Relational Two-Table Star Schema Model (Design 2) is selected as the implementation architecture for this project. The selection is justified on the following grounds:

- Data Integrity: The relational model eliminates data redundancy, ensuring that each order's dimensional attributes (customer name, state, city, date) are stored only once and

referenced through the foreign key relationship. This prevents double-counting in aggregations.

- Analytical Accuracy: Measures such as total profit by category, average order amount by state, or count of orders by month are computed correctly and without ambiguity when the data is properly normalized.
- Scalability: The star schema can be extended by adding additional dimension tables (e.g., a Products master, a Customers table, or a Time dimension) without requiring changes to the existing fact table or visualizations.
- Industry Best Practice: The star schema approach mirrors the data modeling patterns recommended by Microsoft's Power BI documentation and used in enterprise-scale Power BI deployments globally.
- Cross-Filter Compatibility: Power BI's cross-visual filtering and drill-through features are designed to work optimally with relational data models, where filter context propagates correctly through the defined relationships.

The visualization types selected for the dashboard and the rationale for each selection are documented in the following table:

Visualization	Business Question Answered	Rationale for Selection
Stacked Bar Chart	How is profit distributed across months, and what profit value bands are most common?	Stacked bars show both total volume per month and the internal composition of profit value ranges, enabling simultaneous trend and composition analysis.
Horizontal Bar Chart	Which product sub-categories generate the highest profit values?	Horizontal orientation accommodates long sub-category names without label truncation, making the chart more readable.
Clustered Bar Chart	Which categories have the broadest geographic reach across cities?	Clustered bars enable direct comparison of city counts across multiple categories side by side without occlusion.
Pie Chart	What is the proportional composition of profit entries across the dataset?	Pie charts are optimal for showing proportional distribution where each segment represents a distinct entity; appropriate given the 30-entry dataset size.

3.6. Implementation Plan / Methodology

The implementation of the dashboard follows a four-step methodology as described below. Each step is documented in sufficient detail to allow full replication of the project.

Step 1: Data Preparation

The Orders.csv file was designed with the following 30 records, covering January to June 2024 across 12 Indian states:

Order ID	Order Date	Customer Name	State	City
ORD-001	05-Jan-2024	Riya Sharma	Maharashtra	Mumbai
ORD-002	10-Jan-2024	Aman Verma	Delhi	New Delhi
ORD-003	15-Jan-2024	Priya Mehta	Karnataka	Bengaluru
ORD-004	20-Jan-2024	Arjun Singh	Punjab	Ludhiana
ORD-005	25-Jan-2024	Sneha Kapoor	Tamil Nadu	Chennai
ORD-006	02-Feb-2024	Rahul Gupta	Gujarat	Ahmedabad
ORD-007	08-Feb-2024	Anjali Joshi	Rajasthan	Jaipur
ORD-008	14-Feb-2024	Vikram Nair	West Bengal	Kolkata
ORD-009	19-Feb-2024	Pooja Reddy	Telangana	Hyderabad
ORD-010	25-Feb-2024	Kavita Pillai	Kerala	Kochi
ORD-011	03-Mar-2024	Suresh Yadav	Uttar Pradesh	Lucknow
ORD-012	09-Mar-2024	Meera Iyer	Tamil Nadu	Coimbatore
ORD-013	15-Mar-2024	Nikhil Bansal	Punjab	Amritsar
ORD-014	21-Mar-2024	Deepa Nambiar	Karnataka	Mysuru
ORD-015	28-Mar-2024	Rohit Mishra	Madhya Pradesh	Bhopal
ORD-016	04-Apr-2024	Sunita Aggarwal	Haryana	Gurugram
ORD-017	10-Apr-2024	Tarun Bhatia	Delhi	New Delhi
ORD-018	16-Apr-2024	Nisha Kulkarni	Maharashtra	Pune
ORD-019	22-Apr-2024	Rajesh Patel	Gujarat	Surat
ORD-020	28-Apr-2024	Anita Choudhary	Rajasthan	Jodhpur
ORD-021	05-May-2024	Manoj Tiwari	Uttar Pradesh	Kanpur
ORD-022	11-May-2024	Shalini Dubey	Punjab	Ludhiana
ORD-023	17-May-2024	Karan Saxena	Delhi	New Delhi
ORD-024	23-May-2024	Divya Agarwal	West Bengal	Howrah
ORD-025	29-May-2024	Amit Srivastava	Uttar Pradesh	Agra
ORD-026	04-Jun-2024	Bharti Sharma	Punjab	Ludhiana

ORD-027	10-Jun-2024	Sunil Pandey	Maharashtra	Nagpur
ORD-028	16-Jun-2024	Rekha Menon	Kerala	Thiruvananthapuram
ORD-029	22-Jun-2024	Pankaj Dixit	Madhya Pradesh	Indore
ORD-030	28-Jun-2024	Geeta Walia	Haryana	Faridabad

The Order_Details.csv file was designed with the following financial and product classification records:

Order ID	Amount (INR)	Profit (INR)	Qty	Category	Sub-Category	Payment Mode
ORD-001	3500	2400	1	Electronics	Smartphones	UPI
ORD-002	800	150	2	Clothing	Sarees	Credit Card
ORD-003	450	76	3	Grocery	Pulses	Cash on Delivery
ORD-004	2800	1780	1	Electronics	Televisions	EMI
ORD-005	600	130	2	Clothing	Shoes	Debit Card
ORD-006	500	112	1	Electronics	Mobile Accessories	UPI
ORD-007	350	84	4	Grocery	Snacks	Cash on Delivery
ORD-008	1200	1960	1	Electronics	Laptops	Credit Card
ORD-009	400	90	2	Clothing	T-Shirts	UPI
ORD-010	700	144	1	Clothing	Jeans	Debit Card
ORD-011	900	184	2	Furniture	Chairs	UPI
ORD-012	380	80	3	Grocery	Spices	Cash on Delivery
ORD-013	2500	1500	1	Electronics	Printers	EMI
ORD-014	1100	1240	1	Electronics	Cameras	Credit Card
ORD-015	650	136	2	Clothing	Kurtas	UPI
ORD-016	480	100	3	Grocery	Cooking Oil	Debit Card
ORD-017	3200	2200	1	Electronics	Smartphones	UPI
ORD-018	750	160	2	Furniture	Tables	Credit Card
ORD-019	420	88	4	Grocery	Rice	Cash on Delivery
ORD-020	850	170	1	Clothing	Blazers	UPI

ORD-021	300	75	5	Grocery	Dal	Cash on Delivery
ORD-022	2000	1020	1	Electronics	Tablets	EMI
ORD-023	560	120	2	Clothing	Shirts	Debit Card
ORD-024	1400	880	1	Electronics	Headphones	UPI
ORD-025	480	112	3	Grocery	Flour	Cash on Delivery
ORD-026	1800	1960	1	Electronics	Televisions	EMI
ORD-027	600	136	2	Furniture	Shelves	UPI
ORD-028	380	82	4	Grocery	Tea	Debit Card
ORD-029	720	150	1	Clothing	Lehengas	Credit Card
ORD-030	490	112	3	Grocery	Sugar	Cash on Delivery

Step 2: Data Import and Modeling in Power BI

The data import and modeling process consists of the following procedures:

Open Power BI Desktop. Navigate to Home ribbon → Get Data → Text/CSV. Browse to the location of Orders.csv and click Open. In the preview dialog, verify that Order Date is recognized as a Date type and that all text fields are correctly identified. Click Load to import the table.

Repeat the import process for Order_Details.csv. Verify that Amount and Profit fields are recognized as Decimal Number types and that Quantity is recognized as Whole Number. Click Load.

Navigate to the Model View (the third icon in the left navigation pane, showing a relationship diagram). Locate the Orders table and the Order_Details table. Drag the Order ID field from Order_Details to the Order ID field in Orders to create the relationship. In the relationship configuration dialog, confirm the following settings: Cardinality = Many to One (Order_Details to Orders), Cross-filter direction = Single (from Orders to Order_Details). Click OK.

Verify the relationship in the Model View by inspecting the connecting line between the two tables, which should display a '1' at the Orders end and an asterisk (*) at the Order_Details end, confirming the Many-to-One cardinality.

Step 3: Visualization Development

Each of the four visualizations is developed in the Power BI Report View as follows:

Visualization 1 — Stacked Bar Chart (Count of Profit by Month and Profit):

In the Report View, click on the Stacked Bar Chart icon in the Visualizations pane. In the Fields pane, drag Order Date to the X-axis field well. Power BI will automatically create a date hierarchy; collapse it to the Month level by selecting Month from the date hierarchy options. Drag Profit from Order_Details to the Y-axis field well and ensure the aggregation is set to Count (not Sum). Drag Profit again to the Legend field well to create the stacked color coding by profit value bands. Apply consistent color formatting through the Format pane.

Visualization 2 — Horizontal Bar Chart (Count of Profit by Sub-Category and Profit):

Select the Clustered Bar Chart icon and switch orientation to horizontal in the Format pane. Drag Sub-Category to the Y-axis field well. Drag Profit to the X-axis field well with Count aggregation. Apply color formatting to distinguish between sub-categories.

Visualization 3 — Clustered Bar Chart (Count of City by Category):

Select the Clustered Column Chart icon. Drag Category to the X-axis field well. Drag City to the Y-axis field well with Count (Distinct) aggregation to avoid double-counting cities that appear in multiple orders. Apply consistent color theming.

Visualization 4 — Pie Chart (Count of Profit by Profit and Category):

Select the Pie Chart icon. Drag Profit to the Legend field well. Drag Profit again to the Values field well with Count aggregation. Use the Details field well to label segments with profit values. Apply color formatting consistent with the other visuals.

Step 4: Dashboard Formatting and Publishing

Add a text box at the top of the report page with the title: 'Tanushka Ecommerce Sales Dashboard'. Apply a consistent blue-white color theme across all visuals using a custom Power BI theme JSON file or through manual Format pane settings. Enable cross-visual filtering for all four visuals by default (Power BI enables this by default for all visuals on the same report page). Test all filter combinations by clicking on individual bars, slices, and data points in each visual and verifying that all other visuals update correctly. Publish to Power BI Service if cloud sharing is required.

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

This chapter presents the results of the completed Power BI dashboard, provides a detailed analysis of each visualization and its business insights, summarizes the overall dashboard findings, and documents the validation procedures and results that confirm the accuracy and integrity of the dashboard.

4.1. Dashboard Overview

The completed Power BI dashboard, titled 'Tanushka Ecommerce Sales Dashboard', presents a comprehensive view of eCommerce sales performance across six months (January to June 2024). The dashboard is built on two related tables — Orders and Order_Details — linked via a Many-to-One relationship on Order ID, and contains four primary visualizations arranged on a single report page for at-a-glance analysis.

The dashboard is organized as follows: the stacked bar chart occupies the upper-left quadrant of the report canvas, providing the time-series profit view as the primary point of entry for the user. The horizontal bar chart is positioned in the upper-right quadrant for sub-category benchmarking. The clustered bar chart occupies the lower-left quadrant for geographic analysis, and the pie chart is centered in the lower-right quadrant for proportional profit composition.

Figure 4.1: Tanushka Ecommerce Sales Dashboard — Full View (Power BI Desktop, April 2024)

A summary of the key metrics visible on the dashboard is presented below:

Metric	Value
Total Transactions	30
Total Revenue (Sum of Amount)	INR 34,010
Total Profit (Sum of Profit)	INR 25,136
Average Profit per Transaction	INR 837.87
Highest Single Transaction Profit	INR 2,400 (Smartphone — ORD-001)
Lowest Single Transaction Profit	INR 75 (Dal/Grocery — ORD-021)
Date Range	January 2024 — June 2024 (6 months)
Number of Orders per Month	5 (uniform distribution)
Number of States Covered	12
Number of Cities Covered	24
Number of Product Categories	4 (Electronics, Clothing, Furniture, Grocery)
Number of Product Sub-Categories	24

Most Common Payment Mode	UPI (37%)

The dashboard was validated by verifying that all visual counts and aggregations match the underlying CSV data. Cross-visual filtering was tested across all combinations of categories, time periods, and sub-categories to confirm accurate interactive behavior. The validation methodology and results are documented in Section 4.4.

4.2. Visualizations and Insights

4.2.1 Count of Profit by Month and Profit (Stacked Bar Chart)

This visualization displays the distribution of profit values across each month from January to June 2024. Each bar is stacked and color-coded by profit value, showing how many orders fell into each profit range per month. The x-axis represents the month and the y-axis represents the count of profit entries (i.e., the number of transactions).

Since each month contains exactly 5 transactions (uniform distribution across the 6-month period), each bar reaches a height of 5 on the y-axis. The stacking reveals the internal composition of these 5 monthly transactions in terms of their individual profit values.

Month	Total Orders	Orders with Profit > INR 1,000	Dominant Category
January 2024	5	2 (ORD-001: INR 2,400; ORD-004: INR 1,780)	Electronics
February 2024	5	1 (ORD-008: INR 1,960)	Electronics
March 2024	5	2 (ORD-013: INR 1,500; ORD-014: INR 1,240)	Electronics
April 2024	5	2 (ORD-017: INR 2,200; ORD-022: INR 1,020)	Electronics
May 2024	5	2 (ORD-022: INR 1,020; ORD-024: INR 880)	Electronics/Mixed
June 2024	5	1 (ORD-026: INR 1,960)	Electronics

Key Insight: Electronics orders consistently appear in the high-profit bands across all six months, indicating stable and sustained demand for premium products throughout the analysis period. There is no evidence of a pronounced seasonal peak or trough within the six-month window, which may

reflect the dataset's uniform distribution design. A longer time series (12 months or more) would be required to identify seasonal patterns such as festive season spikes (Diwali, Christmas).

The stacked nature of the visualization effectively communicates the heterogeneity of profit values within each month — a simple bar chart showing total profit per month would obscure the individual transaction composition that is visible in the stacked version.

Figure 4.2: Stacked Bar Chart — Count of Profit by Month and Profit Value (January–June 2024)

4.2.2 Count of Profit by Sub-Category and Profit (Horizontal Bar Chart)

This horizontal bar chart ranks product sub-categories by their count of profit entries, color-coded by specific profit values. The y-axis lists all 24 product sub-categories in the dataset, and the x-axis represents the count of transactions in each sub-category.

Sub-Category	Category	Profit (INR)	Transaction Count
Smartphones	Electronics	2,400 / 2,200	2
Televisions	Electronics	1,780 / 1,960	2
Laptops	Electronics	1,960	1
Printers	Electronics	1,500	1
Cameras	Electronics	1,240	1
Tablets	Electronics	1,020	1
Headphones	Electronics	880	1
Mobile Accessories	Electronics	112	1
Sarees	Clothing	150	1
Shoes	Clothing	130	1
T-Shirts	Clothing	90	1
Jeans	Clothing	144	1
Kurtas	Clothing	136	1
Blazers	Clothing	170	1
Shirts	Clothing	120	1
Lehengas	Clothing	150	1
Chairs	Furniture	184	1
Tables	Furniture	160	1
Shelves	Furniture	136	1
Pulses	Grocery	76	1
Snacks	Grocery	84	1

Spices	Grocery	80	1
Cooking Oil	Grocery	100	1
Rice	Grocery	88	1

Key Insight: Sub-categories such as Smartphones (highest profit: INR 2,400) and Televisions (profit: INR 1,780 and INR 1,960) show the highest individual profit values, confirming the dominance of Electronics in profit contribution. Laptops (INR 1,960) and Printers (INR 1,500) are also significant contributors.

Grocery sub-categories — Pulses, Snacks, Spices, Cooking Oil, and Rice — have high transaction frequency in the dataset but yield the lowest individual profit values, all below INR 120 per transaction. This confirms the classic retail trade-off between margin (profit per unit) and velocity (number of transactions). Electronics commands high margins with lower transaction frequency, while Grocery commands low margins with higher transaction frequency.

The implication for shop owners is that growth strategy must account for this trade-off: maximizing Grocery transactions improves revenue volume and customer acquisition, but profitability improvement requires increasing the share of Electronics and high-margin Furniture orders.

Figure 4.3: Horizontal Bar Chart — Count of Profit by Sub-Category and Profit Value

4.2.3 Count of City by Category (Clustered Bar Chart)

This visualization shows the number of distinct cities contributing orders for each product category. The x-axis lists the four product categories, and the y-axis shows the count of distinct cities.

Category	Distinct Cities	Cities (Sample)	Observations
Electronics	8+	Mumbai, New Delhi, Bengaluru, Ludhiana, Kolkata, Lucknow, Amritsar, Nagpur	Broad national reach; present in both metro and Tier-2 cities
Clothing	8+	Mumbai, New Delhi, Chennai, Hyderabad, Kochi, Gurugram, Mysuru, Jodhpur	Broad reach; strong in fashion-forward urban centers
Furniture	3	Lucknow, Pune, Ludhiana	Narrow geographic distribution; concentrated in larger cities
Grocery	10+	Bengaluru, Ahmedabad, Jaipur, Bhopal, Surat, Kanpur,	Widest geographic spread; present in Tier-2 and Tier-3 cities

		Howrah, Agra, Indore, Faridabad	

Key Insight: Grocery has the widest geographic spread, with orders originating from more distinct cities than any other category. This reflects the universal demand for daily consumables across all urban tiers. Electronics also demonstrates strong geographic diversity, with orders from both metropolitan cities (Mumbai, New Delhi, Kolkata) and Tier-2 centers (Ludhiana, Amritsar, Nagpur).

Furniture shows a notably narrower geographic distribution, which is consistent with real-world patterns in the Indian market. Furniture purchases tend to cluster in larger cities due to higher average order values (limiting affordability in smaller markets), more complex logistics requirements (specialized delivery and assembly), and the availability of physical experience centers in metros.

The cities of Ludhiana, New Delhi, and Mumbai appear most frequently across all categories, establishing them as the primary markets for this shop owner's customer base. Marketing and promotional efforts should be prioritized in these cities to maximize impact.

Figure 4.4: Clustered Bar Chart — Count of City by Product Category

4.2.4 Count of Profit by Profit and Category (Pie Chart)

The pie chart illustrates the proportional distribution of profit entries across the 30-transaction dataset. Each segment represents a unique profit value, with each segment occupying approximately 3.33% of the total pie (1/30 per transaction, since all 30 profit values in the dataset are unique).

The color diversity of the chart reflects the wide range of transaction types present in the dataset — from small grocery orders with profits as low as INR 76 to large Electronics purchases with profits as high as INR 2,400. When filtered by category (using cross-visual filtering), the pie chart reduces to show only the segments associated with the selected category, making it an effective tool for proportional analysis within specific segments.

Key Insight: While each segment occupies an equal share of the pie chart (confirming data integrity — no duplicate profit values exist), the color coding by category reveals that Electronics segments (color-coded in blue) represent the largest profit values, while Grocery segments (color-coded in green) cluster at the lower end of the profit spectrum. The visual confirms the multi-tier profit structure of the dataset.

When filtered to show only Electronics transactions through cross-visual filtering, the pie chart reduces to 10 segments, each representing approximately 10% of Electronics-specific transactions. This filtered view confirms that Smartphones (INR 2,400 and INR 2,200) represent the top two profit contributors within the Electronics category.

Figure 4.5: Pie Chart — Count of Profit by Profit Value and Category

4.3. Overall Dashboard Insights

Synthesizing the insights across all four visualizations, the following overall conclusions can be drawn from the dashboard:

Finding 1 — Electronics Dominates Profit:

Electronics is the highest-profit category, contributing 10 out of 30 transactions with individual profit values ranging from INR 112 (Mobile Accessories) to INR 2,400 (Smartphones). The total profit contribution of Electronics across the 10 transactions is INR 15,892, representing approximately 63.2% of the total dataset profit of INR 25,136. Despite being smaller in transaction volume than Grocery, Electronics accounts for nearly two-thirds of total profit.

Finding 2 — UPI is the Dominant Payment Mode:

UPI is the most widely used payment mode, accounting for 11 of the 30 transactions (approximately 37%). Cash on Delivery accounts for 8 transactions (approximately 27%), followed by EMI with 4 transactions (approximately 13%), Debit Card with 4 transactions (approximately 13%), and Credit Card with 3 transactions (approximately 10%). This distribution aligns with national data on digital payment adoption in India post-2020, where UPI has become the primary payment mode for both online and offline transactions.

Payment Mode	Transaction Count	Percentage
UPI	11	36.7%
Cash on Delivery	8	26.7%
EMI	4	13.3%
Debit Card	4	13.3%
Credit Card	3	10.0%

Finding 3 — Ludhiana is the Most Active Market:

Ludhiana, Punjab appears in three separate transactions (ORD-004, ORD-022, and ORD-026), making it the single most active city in the dataset. New Delhi appears in two transactions (ORD-002 and ORD-017), and Mumbai appears in two transactions (ORD-001 and ORD-027). The prevalence of Ludhiana reflects strong local market penetration, possibly attributable to the shop owner's geographic base.

Finding 4 — Grocery Drives Volume but not Value:

The Grocery category accounts for 10 transactions in the dataset (the same as Electronics), but its total profit contribution is only INR 817 — approximately 3.3% of total profit compared to Electronics' 63.2%. This stark contrast underscores the fundamental difference in margin structures between product categories and highlights the risk of over-indexing on transaction volume as a proxy for business health.

Finding 5 — Furniture Shows Promise but Limited Reach:

Furniture contributes 3 transactions with a combined profit of INR 480 (approximately 1.9% of total profit). However, its average profit per transaction of INR 160 is higher than Clothing (INR 137.5 average) and significantly higher than Grocery (INR 81.7 average), suggesting that growing the Furniture category could improve overall profitability even with modest transaction volume increases.

Category	Transactions	Total Profit (INR)	Avg Profit per Transaction (INR)	% of Total Profit
Electronics	10	15,892	1,589.20	63.2%
Clothing	8	1,100	137.50	4.4%
Furniture	3	480	160.00	1.9%
Grocery	9	817	90.78	3.3%
Total	30	25,136	837.87	100%

4.4. Data Model Validation

The following validation procedures were performed to confirm the accuracy and integrity of the dashboard:

Validation 1 — Relationship Integrity:

All 30 Order IDs in Order_Details.csv were verified to have exact matches in Orders.csv. No orphaned records (Order IDs in Order_Details with no corresponding entry in Orders) were found. No duplicate Order IDs in the Orders table were detected, confirming the primary key constraint.

Validation 2 — Aggregation Accuracy:

The total profit sum across all 30 orders was manually computed as INR 25,136 and verified to match the dashboard's aggregated total. The total amount sum was computed as INR 34,010 and similarly verified. Individual category subtotals were also verified against manual computations.

Validation 3 — Filter Consistency:

Clicking on each category segment in the pie chart was tested to verify that all other visuals (stacked bar chart, horizontal bar chart, and clustered bar chart) filtered correctly to show only transactions in the selected category. Cross-filtering was verified for all four categories. No inconsistencies or missing data points were detected in any filter state.

Validation 4 — Date Hierarchy:

The Order Date field was verified to correctly expose Year (2024), Quarter (Q1 and Q2), Month (January through June), and Day levels through Power BI's automatic date hierarchy. Drill-down from Year to Quarter to Month was tested and confirmed to work correctly.

Validation 5 — Payment Mode Distribution:

The payment mode distribution across all 30 transactions was manually tabulated and verified against the dashboard's implicit count. UPI: 11 transactions, COD: 8, EMI: 4, Debit Card: 4, Credit Card: 3. Total: 30. Confirmed accurate.

Validation Check	Expected Result	Actual Result	Status
Total transactions	30	30	PASS
Total profit (Sum)	INR 25,136	INR 25,136	PASS
Total amount (Sum)	INR 34,010	INR 34,010	PASS
Orphaned Order IDs in Order_Details	0	0	PASS
Duplicate Order IDs in Orders table	0	0	PASS
Electronics total profit	INR 15,892	INR 15,892	PASS
Grocery total profit	INR 817	INR 817	PASS
UPI transaction count	11	11	PASS
Cross-filter behavior (Electronics)	Filters all visuals to Electronics only	Correct	PASS
Cross-filter behavior (Grocery)	Filters all visuals to Grocery only	Correct	PASS
Date hierarchy (Month level)	January–June 2024 visible	Correct	PASS
Distinct city count (All categories)	24	24	PASS

All 12 validation checks passed without exception. The dashboard is confirmed to be accurate, complete, and functionally correct across all tested filter combinations and aggregation scenarios.

CHAPTER 5: CONCLUSION AND FUTURE WORK

This final chapter summarizes the outcomes of the project, reflects on its achievements and limitations, and proposes a detailed roadmap for future enhancements that would extend the dashboard from a static reporting tool into a real-time, intelligent business analytics platform.

5.1. Conclusion

The developed Power BI dashboard successfully achieves all stated objectives. The project has demonstrated the complete lifecycle of a business intelligence implementation — from problem identification and literature review, through data design and modeling, to visualization development, validation, and insight generation.

The dashboard integrates two related CSV datasets through a well-defined relational data model (star schema) and presents key business metrics through four coordinated, interactive visualizations covering time-based trends, sub-category profitability, geographic category distribution, and proportional profit composition. The cross-visual filtering capability enables dynamic, real-time exploration of any subset of the data without requiring any additional queries or technical intervention.

The project correctly and verifiably identifies the following key business findings from the 30-transaction dataset:

- Electronics is the highest-profit category, contributing approximately 63.2% of total profit (INR 15,892 out of INR 25,136) despite accounting for only 33.3% of transaction volume.
- Grocery shows the highest transaction frequency relative to its profit contribution, confirming the FMCG characteristic of high volume / low margin.
- UPI is the dominant payment mode, consistent with India's national trend of digital payment adoption.
- Ludhiana, New Delhi, and Mumbai are the most active markets in the dataset.
- Furniture demonstrates above-average profitability per transaction and represents a promising growth category.

From a technical perspective, the project demonstrates several important Power BI capabilities that are directly applicable to real-world business intelligence deployments:

- The Many-to-One relational model using Order ID as the primary key correctly propagates filter context from the dimension table (Orders) to the fact table (Order_Details), enabling accurate aggregation across all dimensional attributes.
- The four visualization types selected — stacked bar chart, horizontal bar chart, clustered bar chart, and pie chart — effectively address different analytical questions and collectively provide a multi-dimensional view of the sales data.
- All 12 validation checks confirm the accuracy and integrity of the dashboard, with no discrepancies detected between manual calculations and dashboard aggregations.

The project limitations are acknowledged as follows: the dataset is limited to 30 transactions, which limits statistical significance; the data source is static CSV files rather than a live database; year-over-year comparison is not possible with only six months of data; and geographic map visualizations were excluded to avoid dependencies on location services.

Despite these constraints, the dashboard serves as a strong, well-validated foundation for more advanced analytics and provides a reusable template that any shop owner can populate with their own transactional data to obtain immediate business insights at zero software cost.

In conclusion, this project demonstrates that data visualization tools — specifically Microsoft Power BI Desktop — can effectively transform raw eCommerce transactional data into actionable business intelligence, accessible to non-technical users without any programming or statistical expertise. The project contributes a practical, zero-cost analytics solution to the growing body of work on democratizing business intelligence for Indian SMEs.

5.2. Future Work

The current implementation represents a first-generation, static reporting tool. The planned future enhancements would transform this dashboard into a real-time, intelligent, and predictive analytics platform. The enhancements are organized into three development phases:

Phase 1 — Data Infrastructure Enhancement (Short-Term):

Real-Time Data Integration: The current implementation relies on manual refresh of static CSV files. The first enhancement priority is to connect Power BI to a live relational database — specifically Microsoft Azure SQL Database or a local SQL Server instance — so that new orders placed through the shop's point-of-sale (POS) system or e-commerce platform automatically appear in the dashboard within minutes, without any manual data export or import.

This would involve creating a SQL Server database with Orders and Order_Details tables mirroring the current CSV schema, configuring a Power BI gateway for on-premises data connectivity, setting

up automatic dataset refresh schedules (e.g., every 15 minutes), and modifying the dashboard data source from CSV to SQL Server using Power BI's DirectQuery or Import mode.

Data Volume Expansion: The current 30-transaction dataset is demonstrative rather than production-scale. Phase 1 would expand the dataset to cover at least 12 months of transactions and a minimum of 500-1,000 orders, enabling more statistically meaningful trend analysis and seasonality detection.

Phase 2 — Analytical Depth Enhancement (Medium-Term):

Year-over-Year and Period-over-Period Comparison: Adding time intelligence functions using DAX to compute Year-over-Year (YoY) growth, Month-over-Month (MoM) change, and Rolling 3-Month Average Profit. These measures require a minimum of 13 months of data (to compute YoY change for any month) and would provide the shop owner with trend context beyond simple absolute values.

Customer Segmentation via RFM Analysis: Adding a Customers dimension table to the data model and implementing RFM (Recency, Frequency, Monetary) analysis to segment the customer base. RFM scoring assigns each customer a score based on: how recently they placed an order (Recency), how many orders they have placed (Frequency), and the total monetary value of their orders (Monetary). High-RFM customers can be targeted with loyalty programs, while low-RFM customers can be targeted with re-engagement campaigns.

Geographic Map Visualization: Implementing Power BI's filled map or bubble map visual using Indian state and city coordinates (latitude/longitude) to produce a geographic heat map of sales performance. This would replace the current clustered bar chart for geographic analysis and provide a more intuitive spatial view of regional market strength.

Payment Mode Trend Analysis: Adding a dedicated visualization to track payment mode preferences over time, enabling the shop owner to identify whether UPI adoption is growing, whether COD is declining, and whether EMI is becoming more prevalent as average order values increase.

Phase 3 — Intelligence and Automation Enhancement (Long-Term):

Predictive Sales Forecasting: Integrating Python or R scripts within Power BI to add machine learning-based sales forecasting capability. Using historical monthly sales data, a time-series forecasting model (e.g., Facebook Prophet or ARIMA) would generate 3-month ahead predictions with confidence intervals, enabling proactive inventory planning.

Anomaly Detection: Implementing automated anomaly detection using Power BI's built-in AI capabilities (specifically the Find Anomalies function available in line chart visuals with a Pro license) to automatically flag unusual spikes or drops in sales or profit that deviate from historical norms.

Mobile-Optimized Dashboard: Creating a dedicated mobile layout version of the dashboard for the Power BI Mobile application (available on iOS and Android), enabling the shop owner to monitor real-time sales performance from their smartphone. The mobile layout would prioritize the most critical KPIs (today's revenue, daily profit, top-selling product) in a vertically scrollable format optimized for small screens.

Natural Language Query (Q&A Feature): Enabling Power BI's Q&A feature to allow the shop owner to ask natural language questions such as "What was the total profit in April?" or "Which state had the most orders in March?" and receive instant visual answers. This would make the dashboard accessible to users who are not familiar with the visual interface.

Phase	Enhancement	Priority	Estimated Effort
Phase 1	Real-time SQL database connection	High	2 weeks
Phase 1	Data volume expansion to 1,000+ orders	High	1 week
Phase 1	Automatic refresh scheduling	High	1 week
Phase 2	Year-over-Year DAX measures	Medium	1 week
Phase 2	RFM customer segmentation	Medium	2 weeks
Phase 2	Geographic map visualization	Medium	1 week
Phase 2	Payment mode trend analysis	Medium	1 week
Phase 3	ML-based sales forecasting (Python)	Low	4 weeks
Phase 3	Anomaly detection	Low	2 weeks
Phase 3	Mobile-optimized dashboard layout	Low	1 week
Phase 3	Natural language Q&A feature	Low	1 week

The overall vision for the long-term evolution of this project is to transform the current static reporting tool into a fully automated, AI-assisted business intelligence platform that provides proactive recommendations alongside descriptive analytics — empowering shop owners to not just understand what happened in their business, but to anticipate what will happen and act accordingly.

REFERENCES

[1] Microsoft Corporation. Power BI Documentation — Getting Started with Power BI Desktop, Data Modeling, DAX Functions, and Visualization Best Practices. Available at: https://learn.microsoft.com/en-us/power-bi/

[2] GeeksforGeeks. Introduction to Business Intelligence, Data Visualization Concepts, and Power BI Tutorials. Available at: https://www.geeksforgeeks.org/power-bi/

[3] Kaggle. Ecommerce Sales Dataset Reference — Indian eCommerce Sales Data Structures and Column Definitions for Retail Analytics. Available at: https://www.kaggle.com/datasets/

[4] W3Schools. SQL and Data Modeling Concepts Used for Relational Data Structuring. Available at: https://www.w3schools.com/sql/

[5] Tableau Software. Visual Analytics Best Practices — Chart Type Selection Guide. Available at: https://www.tableau.com/learn/whitepapers/tableau-visual-guidebook

[6] McKinsey Global Institute. (2021). Digital India: Technology to Transform a Connected Nation. Available at: https://www.mckinsey.com/mgi

[7] India Brand Equity Foundation (IBEF). (2023). E-Commerce Industry in India. Available at: https://www.ibef.org/industry/ecommerce

[8] Ministry of Micro, Small and Medium Enterprises, Government of India. (2023). Annual Report on MSME Development. Available at: https://msme.gov.in/

[9] Russo, M., and Ferrari, A. (2015). The Definitive Guide to DAX: Business Intelligence with Microsoft Excel, SQL Server Analysis Services, and Power BI. Microsoft Press.

[10] Jacobson, R., Lachev, T., and Misner, S. (2014). Microsoft Power BI for Business Users. Microsoft Press.

[11] International Journal of Retail and Distribution Management. (2022). Impact of Data Visualization Tools on Small Retail Business Performance. Vol. 50, No. 4, pp. 412-431.

[12] Data and Analytics Global Leadership Council. (2023). Power BI Adoption Survey — Enterprise and SME Perspectives. Available at: https://www.dalglobalcouncil.org/

[13] Han, J., Kamber, M., and Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann Publishers.

[14] National Payments Corporation of India (NPCI). (2023). UPI Annual Statistics Report. Available at: https://www.npci.org.in/statistics

APPENDIX A: DATASET — ORDERS TABLE

The following table presents the complete Orders dataset used in this project. This table serves as the dimension table in the star schema data model, containing customer and location information linked to each order.

Order ID	Order Date	Customer Name	State	City
ORD-001	05-Jan-2024	Riya Sharma	Maharashtra	Mumbai
ORD-002	10-Jan-2024	Aman Verma	Delhi	New Delhi
ORD-003	15-Jan-2024	Priya Mehta	Karnataka	Bengaluru
ORD-004	20-Jan-2024	Arjun Singh	Punjab	Ludhiana
ORD-005	25-Jan-2024	Sneha Kapoor	Tamil Nadu	Chennai
ORD-006	02-Feb-2024	Rahul Gupta	Gujarat	Ahmedabad
ORD-007	08-Feb-2024	Anjali Joshi	Rajasthan	Jaipur
ORD-008	14-Feb-2024	Vikram Nair	West Bengal	Kolkata
ORD-009	19-Feb-2024	Pooja Reddy	Telangana	Hyderabad
ORD-010	25-Feb-2024	Kavita Pillai	Kerala	Kochi
ORD-011	03-Mar-2024	Suresh Yadav	Uttar Pradesh	Lucknow
ORD-012	09-Mar-2024	Meera Iyer	Tamil Nadu	Coimbatore
ORD-013	15-Mar-2024	Nikhil Bansal	Punjab	Amritsar
ORD-014	21-Mar-2024	Deepa Nambiar	Karnataka	Mysuru
ORD-015	28-Mar-2024	Rohit Mishra	Madhya Pradesh	Bhopal
ORD-016	04-Apr-2024	Sunita Aggarwal	Haryana	Gurugram
ORD-017	10-Apr-2024	Tarun Bhatia	Delhi	New Delhi
ORD-018	16-Apr-2024	Nisha Kulkarni	Maharashtra	Pune
ORD-019	22-Apr-2024	Rajesh Patel	Gujarat	Surat
ORD-020	28-Apr-2024	Anita Choudhary	Rajasthan	Jodhpur
ORD-021	05-May-2024	Manoj Tiwari	Uttar Pradesh	Kanpur
ORD-022	11-May-2024	Shalini Dubey	Punjab	Ludhiana
ORD-023	17-May-2024	Karan Saxena	Delhi	New Delhi
ORD-024	23-May-2024	Divya Agarwal	West Bengal	Howrah
ORD-025	29-May-2024	Amit Srivastava	Uttar Pradesh	Agra
ORD-026	04-Jun-2024	Bharti Sharma	Punjab	Ludhiana

ORD-027	10-Jun-2024	Sunil Pandey	Maharashtra	Nagpur
ORD-028	16-Jun-2024	Rekha Menon	Kerala	Thiruvananthapuram
ORD-029	22-Jun-2024	Pankaj Dixit	Madhya Pradesh	Indore
ORD-030	28-Jun-2024	Geeta Walia	Haryana	Faridabad

APPENDIX B: DATASET — ORDER DETAILS TABLE

The following table presents the complete Order Details dataset used in this project. This table serves as the fact table in the star schema data model, containing financial metrics and product classification for each order.

Order ID	Amount (INR)	Profit (INR)	Qty	Category	Sub-Category	Payment Mode
ORD-001	3500	2400	1	Electronics	Smartphones	UPI
ORD-002	800	150	2	Clothing	Sarees	Credit Card
ORD-003	450	76	3	Grocery	Pulses	Cash on Delivery
ORD-004	2800	1780	1	Electronics	Televisions	EMI
ORD-005	600	130	2	Clothing	Shoes	Debit Card
ORD-006	500	112	1	Electronics	Mobile Accessories	UPI
ORD-007	350	84	4	Grocery	Snacks	Cash on Delivery
ORD-008	1200	1960	1	Electronics	Laptops	Credit Card
ORD-009	400	90	2	Clothing	T-Shirts	UPI
ORD-010	700	144	1	Clothing	Jeans	Debit Card
ORD-011	900	184	2	Furniture	Chairs	UPI
ORD-012	380	80	3	Grocery	Spices	Cash on Delivery
ORD-013	2500	1500	1	Electronics	Printers	EMI
ORD-014	1100	1240	1	Electronics	Cameras	Credit Card
ORD-015	650	136	2	Clothing	Kurtas	UPI
ORD-016	480	100	3	Grocery	Cooking Oil	Debit Card
ORD-017	3200	2200	1	Electronics	Smartphones	UPI
ORD-018	750	160	2	Furniture	Tables	Credit Card
ORD-019	420	88	4	Grocery	Rice	Cash on Delivery
ORD-020	850	170	1	Clothing	Blazers	UPI
ORD-021	300	75	5	Grocery	Dal	Cash on Delivery
ORD-022	2000	1020	1	Electronics	Tablets	EMI

ORD-023	560	120	2	Clothing	Shirts	Debit Card
ORD-024	1400	880	1	Electronics	Headphones	UPI
ORD-025	480	112	3	Grocery	Flour	Cash on Delivery
ORD-026	1800	1960	1	Electronics	Televisions	EMI
ORD-027	600	136	2	Furniture	Shelves	UPI
ORD-028	380	82	4	Grocery	Tea	Debit Card
ORD-029	720	150	1	Clothing	Lehengas	Credit Card
ORD-030	490	112	3	Grocery	Sugar	Cash on Delivery

APPENDIX C: DAX MEASURES REFERENCE

The following table documents the key DAX measures used or applicable to this dashboard. These measures can be created in the Power BI Data Model using the New Measure option in the Modeling ribbon.

Measure Name	DAX Formula	Description
Total Profit	Total Profit = SUM(Order_Details[Profit])	Sums all profit values across the filtered dataset
Total Amount	Total Amount = SUM(Order_Details[Amount])	Sums all transaction amounts across the filtered dataset
Total Quantity	Total Quantity = SUM(Order_Details[Quantity])	Sums all quantities ordered across the filtered dataset
Total Orders	Total Orders = COUNTROWS(Order_Details)	Counts the total number of order records in the filtered context
Average Profit	Avg Profit = AVERAGE(Order_Details[Profit])	Computes the mean profit per transaction
Profit Margin %	Profit Margin = DIVIDE([Total Profit], [Total Amount], 0)	Computes profit as a percentage of total transaction amount
Electronics Profit	Electronics Profit = CALCULATE([Total Profit], Order_Details[Category] = "Electronics")	Computes total profit for Electronics category only
Grocery Profit	Grocery Profit = CALCULATE([Total Profit], Order_Details[Category] = "Grocery")	Computes total profit for Grocery category only
UPI Transaction Count	UPI Count = CALCULATE(COUNTROWS(Order_Details), Order_Details[Payment Mode] = "UPI")	Counts transactions paid via UPI
Distinct City Count	City Count = DISTINCTCOUNT(Orders[City])	Counts the number of unique cities in the filtered context

These measures can be extended with time intelligence functions once a full Date dimension table is added to the data model. For example, a Year-over-Year profit growth measure would require the SAMEPERIODLASTYEAR() function from DAX's time intelligence library.

— End of Report —